Instituto Superior de Engenharia do Porto

LETI-BDAMD Project

Steinway DataMart ETL

Technical Report

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Introduction

1.1 Context and Objectives

This project aims to develop a Data Mart for "Steinway," a company specializing in the sale of musical instruments and equipment. The current operational system (OLTP) possesses limited capabilities regarding data analysis. Consequently, the primary objective is to build a decision support system that allows for historical analysis of sales made to customers.

The project involves starting from Dimensional Modeling (following Kimball's methodology) to the implementation of Extraction, Transformation, and Loading (ETL) processes using Visual Studio Integration Services (SSIS).

1.2 Requirements and Scope

The development was constrained by specific business rules and technical requirements:

- The Data Mart must support analysis at the most elementary granularity level.
- Data quality issues, such as missing values or domain violations (e.g., invalid Taxpayer Numbers), must be filtered and rejected into specific tables in the Staging Area.
- The loading process must be incremental, capable of updating existing records and inserting new ones without full reloads.
- The final SSIS project must be deployable to the path `C:\Temp\TrabPrat`.

Chapter 2

Dimensional Modeling

2.1 Business Process and Granularity

The selected business process for analysis is **Sales**. To ensure maximum flexibility for future queries, the granularity was defined as **one row per product line per invoice**.

2.2 Star Schema Design

The resulting model consists of a central Fact table and four dimensions.

FactSales Contains the quantitative metrics: *Quantity*, *UnitPrice*, and *TotalLineAmount*.

DimDate A dimensional table essential for temporal analysis.

DimCustomer Derived from the *Customers* and *CustomerTypes*.

DimEmployee Derived from *Employees* and *Departments*.

DimProduct Derived from *Products* and *Families*.

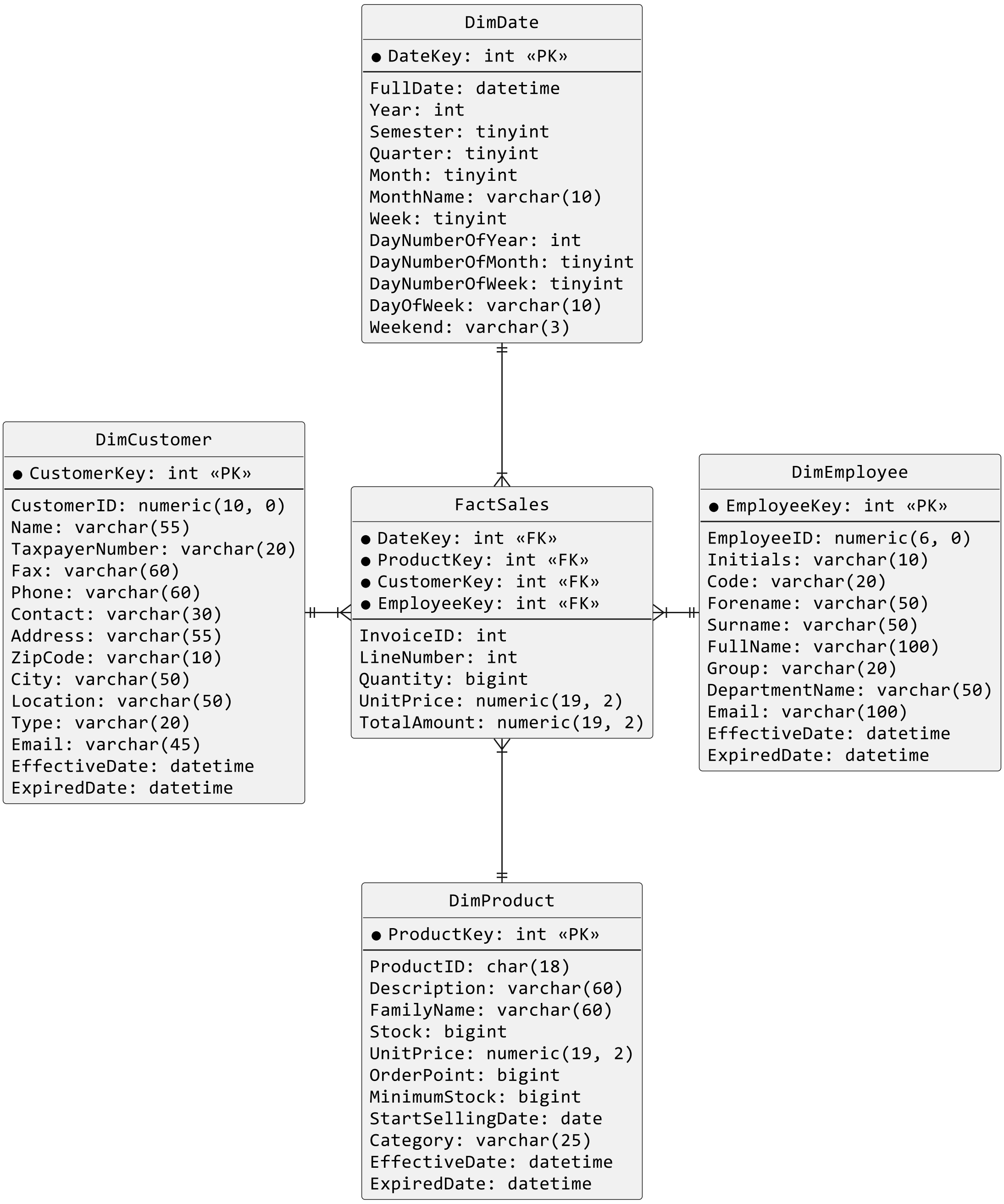

Figure 2.1: Dimensional Model (Star Schema)

2.2.1 Future-Proofing the Model

During the analysis of the source system, additional attributes were identified and added to the dimensions to support potential future requirements, even if not explicitly requested in the initial brief:

- **DimEmployee:** A 'FullName' attribute was created by concatenating names to facilitate reporting.
- **DimProduct:** A 'StockLeftToOrder' attribute was preserved to allow analysis of stock levels versus sales velocity and re-stock urgency.

- **Surrogate Keys:** All dimensions utilize integer Surrogate Keys (e.g., ‘EmployeeID‘) distinct from the source Business Keys.

Chapter 3

ETL Architecture & Implementation

The ETL process was implemented using an SSIS package. The architecture is divided into two main stages: Staging Area population and Data Mart loading.

3.1 Stage 1: Dynamic Staging Area

Instead of hardcoding the creation of staging tables, an automated approach was adopted. A **Foreach Loop Container** iterates through a directory of SQL scripts (‘SQLScripts/Staging‘). This ensures that the Staging Area can be dropped and recreated dynamically on every run, ensuring a clean state before data extraction.

3.2 Stage 2: Transformation and Loading

The transformation phase handles the logic required to convert operational data into analytical information.

3.2.1 Handling the "Departaments" Inconsistency

A significant challenge encountered was a schema inconsistency in the source system. The source table was named ‘Departaments‘ (incorrect spelling), while the target convention for the project was ‘Departments‘. Since the automated script reader used the file name to determine the target table, this caused a conflict.

To resolve this without altering the source system (which simulates a real-world read-only constraint), a dynamic SSIS Expression was utilized within the Control Flow:

```
"IF EXISTS (SELECT * FROM Steinway.sys.tables WHERE name = '" + @[User::
   TableName] + "')
 BEGIN
    INSERT INTO [Steinway_Staging].[dbo].[" + REPLACE(@[User::TableName
   ], "Departaments", "Departments") + "]
    SELECT * FROM [Steinway].[dbo].[" + @[User::TableName] + "]
END"
```

Listing 3.1: Dynamic SQL for Table Renaming

This expression checks the source table name and dynamically replaces the string "Departaments" with "Departments" for the destination, ensuring data flows correctly despite the source typo.

3.2.2 Date Dimension Strategy

The ‘DimDate‘ table required attributes (Semesters, Weekend flags) not present in the source system. A hybrid approach was used: 1. An Analysis Services project generated a comprehensive Date structure. 2. This structure was exported to a CSV file (‘DimDate.csv‘). 3. The SSIS package reads this CSV, performs necessary data type casting (specifically String to Integer to match the Data Mart schema), and loads the ‘DimDate‘ table.

3.2.3 Fact Table Loading and Validation

The loading of ‘FactSales‘ depends on the successful population of all dimensions to ensure referential integrity (Lookup of the Keys). To manage execution dependencies within the SSIS package, the **DelayValidation** property was set to ’`True`’ on the Data Flow Tasks.

This was necessary because the destination tables in the Data Mart might not exist at the very start of the package validation phase (since they are created dynamically by the script execution), which would otherwise cause the package to fail validation before execution began.

3.3 Data Quality and Incremental Loading

As per requirements:

- **Data Cleaning:** Records with domain violations (e.g., ‘TaxpayerNumber‘ = "000000000") are filtered using Conditional Split components and redirected to error tables in the Staging Area.
- **Incremental Load:** ‘DimProduct‘ implements Slowly Changing Dimensions (SCD Type 2) to track historical changes in families and categories, while ‘FactSales‘ loads new invoices based on date comparisons.

Chapter 4

Conclusion

The project successfully delivered a functional ETL pipeline that transforms raw transactional data into an analytical Data Mart. The solution is robust, featuring automated Staging creation and handling complex business requirements like historical tracking of product categories.

The main technical challenges, such as the source schema naming errors and the strict data typing of SSIS, were overcome through dynamic expressions and careful configuration of the Control Flow. The final Star Schema effectively answers the business questions regarding sales performance across time, geography, and product lines.